

PoreBoostGAN :
AI-Supercharged Swin Transformer
for Unprecedented Detail

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Introduction

Super-resolution is an advanced deep learning technique that aims to establish a mapping relationship between the low-frequency signals (contours) of low-resolution images and the high-frequency signals (details) of their high-resolution counterparts. A novel approach that combines Generative Adversarial Networks (GANs) with the SwinIR architecture, which utilizes a hierarchical Swin Transformer based on the attention mechanism. This innovative framework demonstrates excellent adaptability to long-distance-dependent super-resolution tasks while maintaining a compact model size and reduced training complexity.

Tracking changes in rock
properties across scales
and predicting fine
structure beyond the
detection limitation.

Dataset

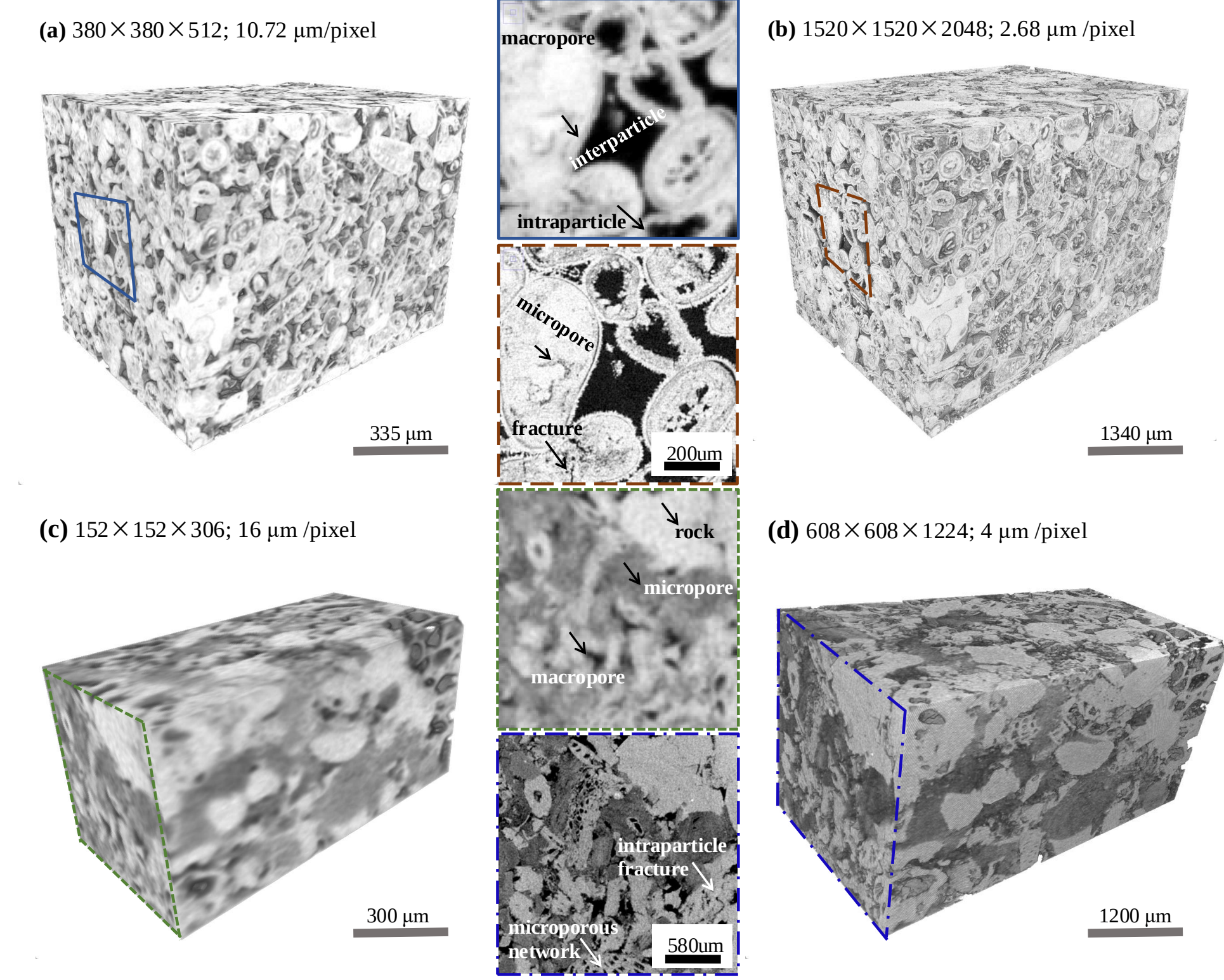


Figure 1: From (a) to (b) denotes a four-fold enhancement of core resolution in the MRCCM dataset, from 10.72μm/pixel to 2.68μm/pixel. (c) to (d) denotes the more challenging biogenic carbonate dataset, which is challenging in that the distribution of features is no longer as uniform as in the MRCCM dataset, and that the cores exhibit a very high degree of spatial inhomogeneity.

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Methodology

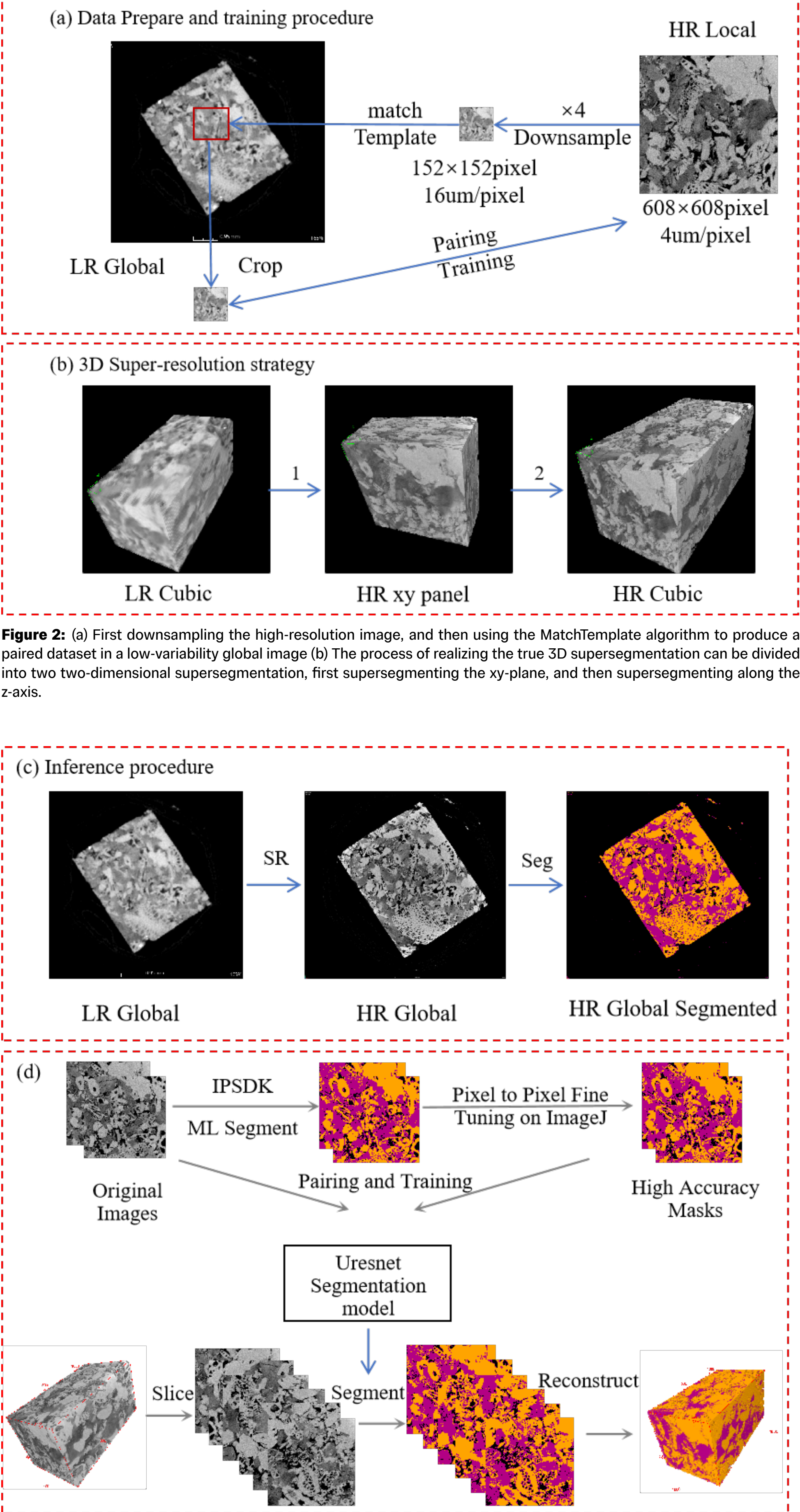


Figure 2: (a) First downsampling the high-resolution image, and then using the MatchTemplate algorithm to produce a paired dataset in a low-variability global image (b) The process of realizing the true 3D supersegmentation can be divided into two two-dimensional supersegmentation, first supersegmenting the xy-plane, and then supersegmenting along the z-axis.

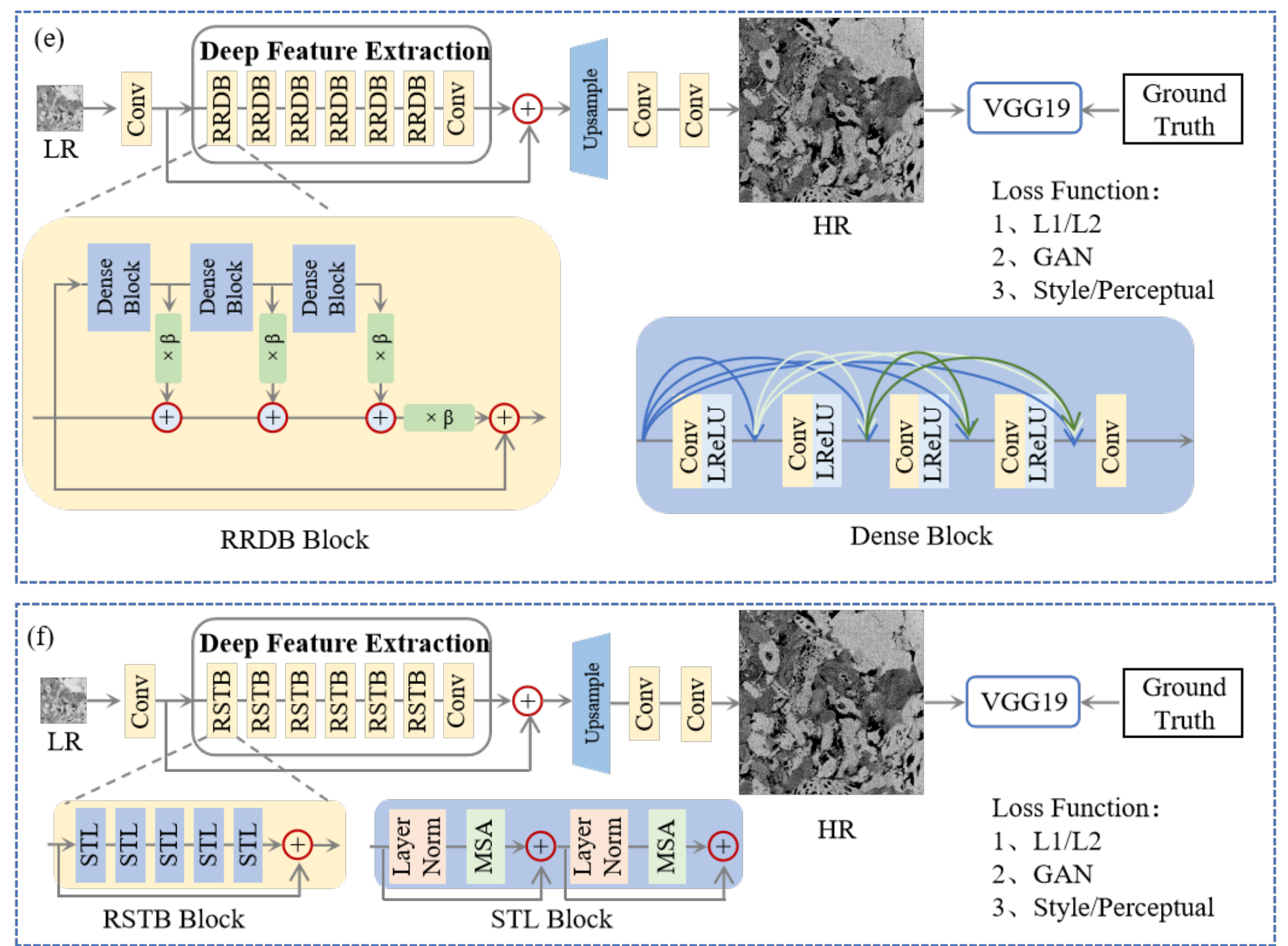


Figure 4: (e) Architecture of the RRDB net. (f) Architecture of the Swin Transformer net.

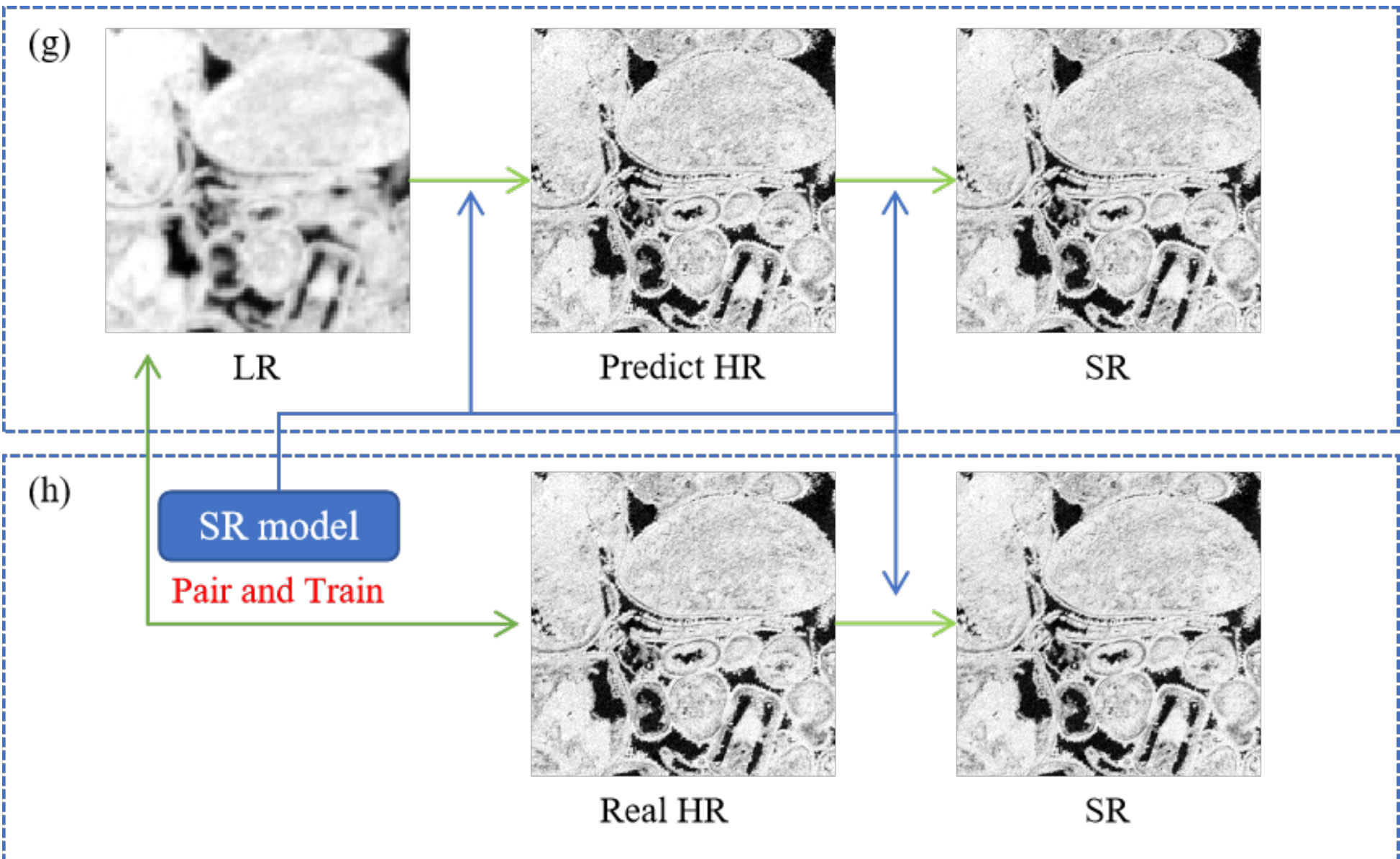


Figure 5: Pairwise training of high and low resolution images to obtain an upsampling model, in order to obtain a resolution that is not achievable with current technology, two promotions are made: 1. Feed the low resolution image into the model twice. 2. Feed the high resolution image into the model

Conclusion

1. Spatial invariance of convolution -> train small datasets apply to arbitrary sizes.
2. Transformer long-range invariance -> complex rock feature models remain stable.
3. The model can understand the pattern of feature representation and change in images at different scales. This understanding can help the model to extrapolate the learnt cross-scale dependencies to higher resolutions in the inference stage.

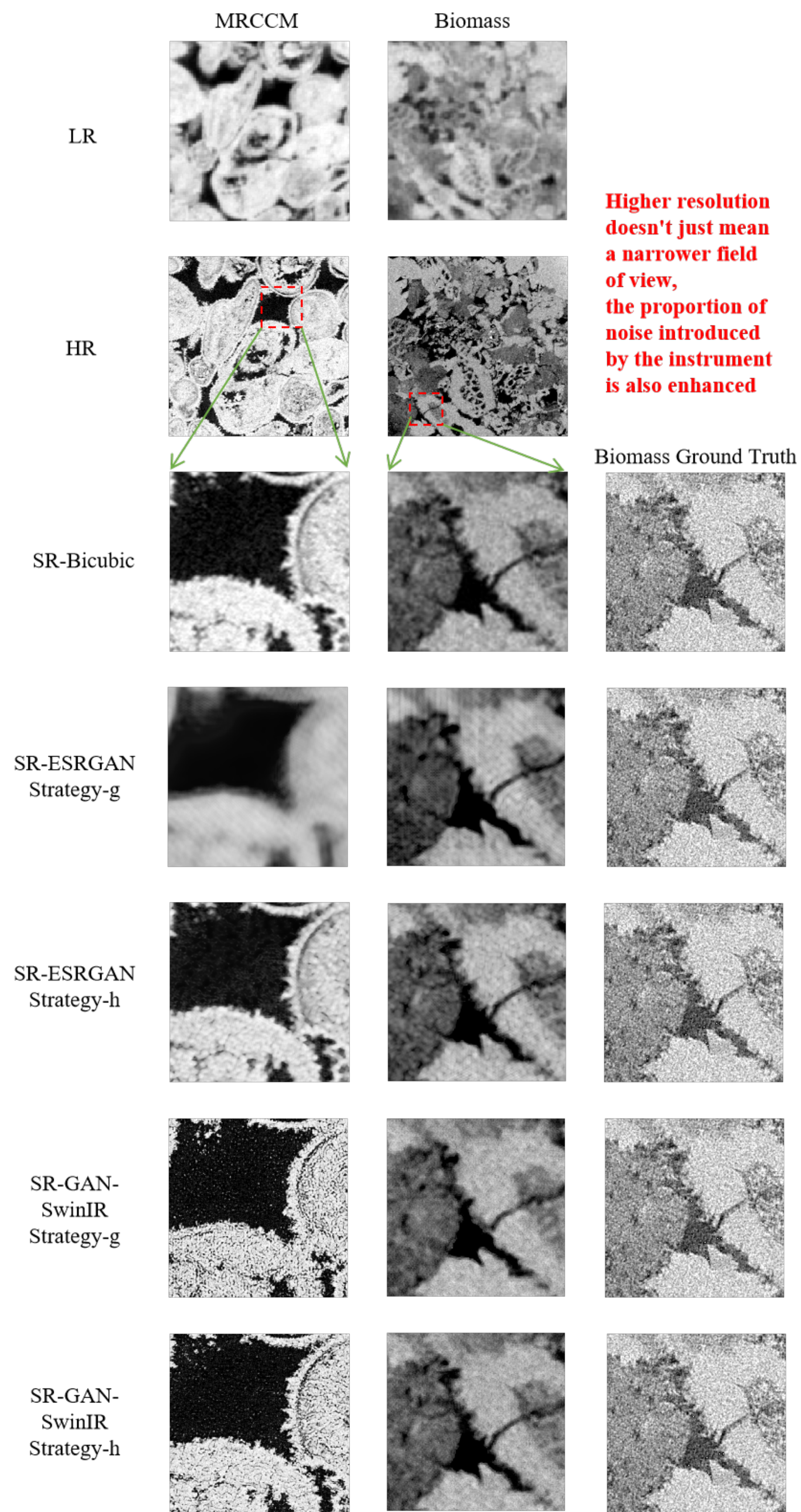


Figure 6: The slices of the two datasets were plotted with two methods to realize the effect of super-resolution, and the biomass carbonate rock dataset has three resolutions of 1, 4, and 16, which provides the basis for the validation of super-resolution reference

Table 1: Physical and statistical parameter improvements of EDSR baseline vs GAN assist SwinIR on biomass dataset (Porosity, Specific Surface Area, Euler Number) show the difference from the ground truth

Parameters	EDSR	SwinIR
SSIM↑	0.46	0.42
PSNR↑	18.29	16.36
NIQE↓	73.97	27.30
PIQE↓	18.99	9.95
Porosity	-39.63%	-2.74%
SSA	-4.32%	+0.55%
Euler	+25.98%	+0.41%